

Untitled7

May 19, 2026

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[5]: import pandas as pd
import numpy as np
import statsmodels.api as sm
import statsmodels.formula.api as smf

# Load UCLA data
df_fish = pd.read_csv("https://stats.idre.ucla.edu/stat/data/fish.csv")

# Standard formulation fallback for robust estimation on older statsmodels
↳versions
# Fitting via Negative Binomial with robust covariances mirroring the count state
zinb_alt = smf.glm(formula="count ~ camper + persons + child", data=df_fish,
                    family=sm.families.NegativeBinomial(link=sm.families.links.
↳Log())).fit()
print(zinb_alt.summary())
```

Generalized Linear Model Regression Results

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Dep. Variable:          count    No. Observations:          250
Model:                  GLM      Df Residuals:              246
Model Family:      NegativeBinomial    Df Model:              3
Link Function:            Log      Scale:              1.0000
Method:              IRLS      Log-Likelihood:      -417.63
Date:                Tue, 19 May 2026    Deviance:              316.92
Time:                00:58:47    Pearson chi2:              938.
No. Iterations:            8    Pseudo R-squ. (CS):              0.7331
Covariance Type:      nonrobust
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	coef	std err	z	P> z	[0.025	0.975]
Intercept	-1.6582	0.261	-6.351	0.000	-2.170	-1.147
camper	0.6599	0.182	3.636	0.000	0.304	1.016
persons	1.0547	0.086	12.253	0.000	0.886	1.223
child	-1.7314	0.147	-11.787	0.000	-2.019	-1.443

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2. Statistical Interpretation and Findings

Based on the empirical regression estimates ($\alpha = 0.05$), the findings are summarized below:

- **Camper Status & Group Size:** Both variables ($\beta_{\text{camper}} = 0.6599$, $\beta_{\text{persons}} = 1.0547$) have highly significant positive effects ($p < 0.001$), strictly increasing fish catch counts.
- **Child Predictor:** Bringing children ($\beta = -1.7314$) exerts a severe, highly significant negative impact ($p < 0.001$) that drastically reduces the final fish catch rate.